Just xgboost

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libraries/etc

```
library(tidymodels)
```

```
## -- Attaching packages --------------------------------------- tidymodels 1.2.0 --

## v broom        1.0.7     v recipes      1.1.0
## v dials        1.3.0     v rsample      1.2.1
## v dplyr        1.1.4     v tibble       3.2.1
## v ggplot2      3.5.1     v tidyr        1.3.1
## v infer        1.0.7     v tune         1.2.1
## v modeldata    1.4.0     v workflows    1.1.4
## v parsnip      1.2.1     v workflowsets 1.1.0
## v purrr        1.0.2     v yardstick    1.3.1

## -- Conflicts ------------------------------------------ tidymodels_conflicts() --
## x purrr::discard() masks scales::discard()
## x dplyr::filter()  masks stats::filter()
## x dplyr::lag()     masks stats::lag()
## x recipes::step()  masks stats::step()
## * Search for functions across packages at https://www.tidymodels.org/find/
```

```
library(xgboost)
```

```
## Warning: package 'xgboost' was built under R version 4.4.2
```

```
##
## Attaching package: 'xgboost'
```

```
## The following object is masked from 'package:dplyr':
##
##     slice
```

```
library(vip)
```

```
## Warning: package 'vip' was built under R version 4.4.2
```

```
##
## Attaching package: 'vip'

## The following object is masked from 'package:utils':
##
##     vi
```

```
library(ROCR)
```

```
## Warning: package 'ROCR' was built under R version 4.4.2
```

```
options(scipen = 4)
```

```
## Data Processing
# Initial data import
customer_data <- read.csv("~/00 Merrimack/DSE6111/Course Project/customer_data_1.csv", header = TRUE)

customer_data$Attrition_Flag <- ifelse(customer_data$Attrition_Flag == "Attrited Customer", 1, 0)
customer_data$Attrition_Flag <- factor(customer_data$Attrition_Flag)

# check levels
levels(customer_data$Attrition_Flag)
```

```
## [1] "0" "1"
```

```
# set seed
set.seed(10)

# get rid of ClientNum identifier
customer_data <- customer_data[ -c(1) ]

# create data split
data_split <- initial_split(customer_data, strata = "Attrition_Flag", prop = 0.75)

# create training and test sets
training_set <- training(data_split)
test_set  <- testing(data_split)

# row count per set
nrow(training_set)
```

```
## [1] 7595
```

```
nrow(test_set)
```

```
## [1] 2532
```

```
# use cv folds and tune
cv_set <- vfold_cv(training_set, v = 10)
```

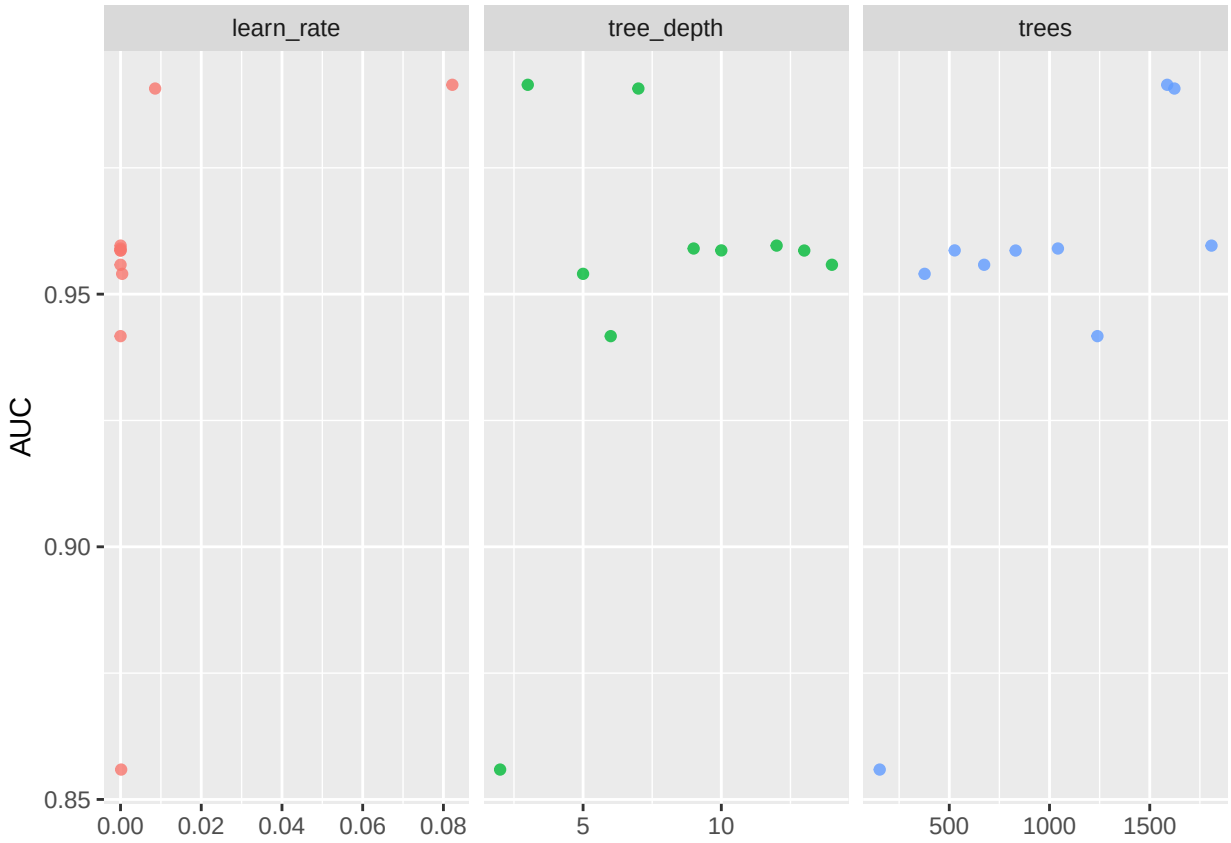
```r
# recipe for preprocessing - only numeric var at first
attrition_recipe_xb <-
  recipe(
    Attrition_Flag ~ Months_on_book + Total_Relationship_Count + Months_Inactive_12_mon + Contacts_Coun
      Total_Amt_Chng_Q4_Q1 + Total_Trans_Amt + Total_Trans_Ct + Total_Ct_Chng_Q4_Q1 +
      Avg_Utilization_Ratio,
    data = training_set
  ) %>%
  step_dummy(all_nominal_predictors()) %>%
  step_center(all_predictors()) %>%
  step_scale(all_predictors())

# model with xgboost
xgboost_model <- boost_tree(
      trees = tune(),
      tree_depth = tune(),
      learn_rate = tune()) %>%
  set_engine("xgboost") %>%
  set_mode("classification")

# create workflow
xgboost_workflow <-
  workflow() %>%
  add_recipe(attrition_recipe_xb) %>%
  add_model(xgboost_model)
```

```r
#################### use tune_grid to find best hyperparameter values ######################

xgboost_grid <- grid_latin_hypercube(
  trees(),
  tree_depth(),
  learn_rate(),
  size = 10
)
```

```
## Warning: `grid_latin_hypercube()` was deprecated in dials 1.3.0.
## i Please use `grid_space_filling()` instead.
## This warning is displayed once every 8 hours.
## Call `lifecycle::last_lifecycle_warnings()` to see where this warning was
## generated.
```

```r
xgboost_results <- xgboost_workflow %>%
  tune_grid(resamples = cv_set,
            grid = xgboost_grid,
            control = control_grid(save_pred = TRUE),
            metrics = metric_set(roc_auc, accuracy))

xgboost_results %>%
  collect_metrics() %>%
  filter(.metric == "roc_auc") %>%
  select(mean, trees:learn_rate) %>%
  pivot_longer(trees:learn_rate,
               values_to = "value",
```

```
            names_to = "parameter"
  ) %>%
  ggplot(aes(value, mean, color = parameter)) +
  geom_point(alpha = 0.8, show.legend = FALSE) +
  facet_wrap(~parameter, scales = "free_x") +
  labs(x = NULL, y = "AUC")
```

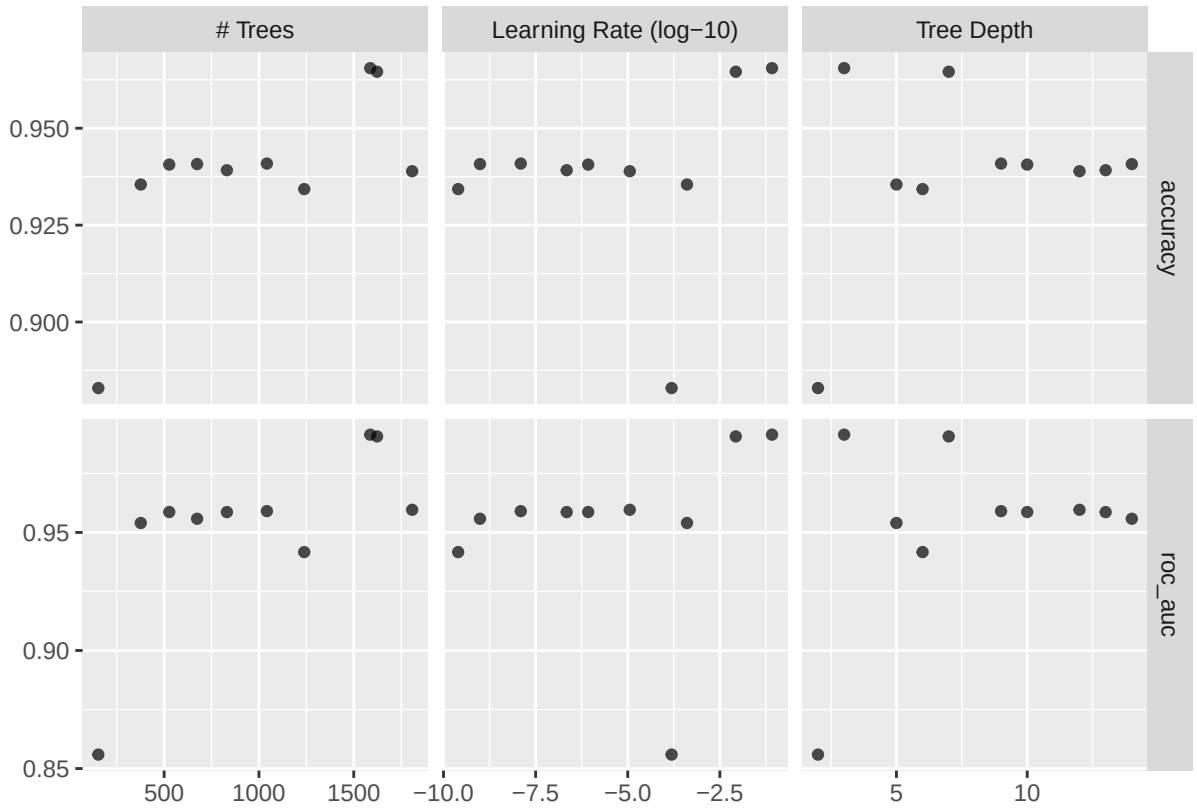



```
# ^ took 30 min to tune 10 models on professor's home computer
```

```
xgboost_results %>% show_best(metric = "roc_auc")
```

```
## # A tibble: 5 x 9
##   trees tree_depth   learn_rate .metric .estimator  mean     n  std_err .config
##   <int>      <int>        <dbl> <chr>   <chr>      <dbl> <int>    <dbl> <chr>
## 1  1587          3 0.0822       roc_auc binary     0.991    10 0.000811 Preproc~
## 2  1623          7 0.00857      roc_auc binary     0.991    10 0.000729 Preproc~
## 3  1808         12 0.0000114    roc_auc binary     0.960    10 0.00351  Preproc~
## 4  1042          9 0.0000000125 roc_auc binary     0.959    10 0.00330  Preproc~
## 5   527         10 0.000000844  roc_auc binary     0.959    10 0.00358  Preproc~
```

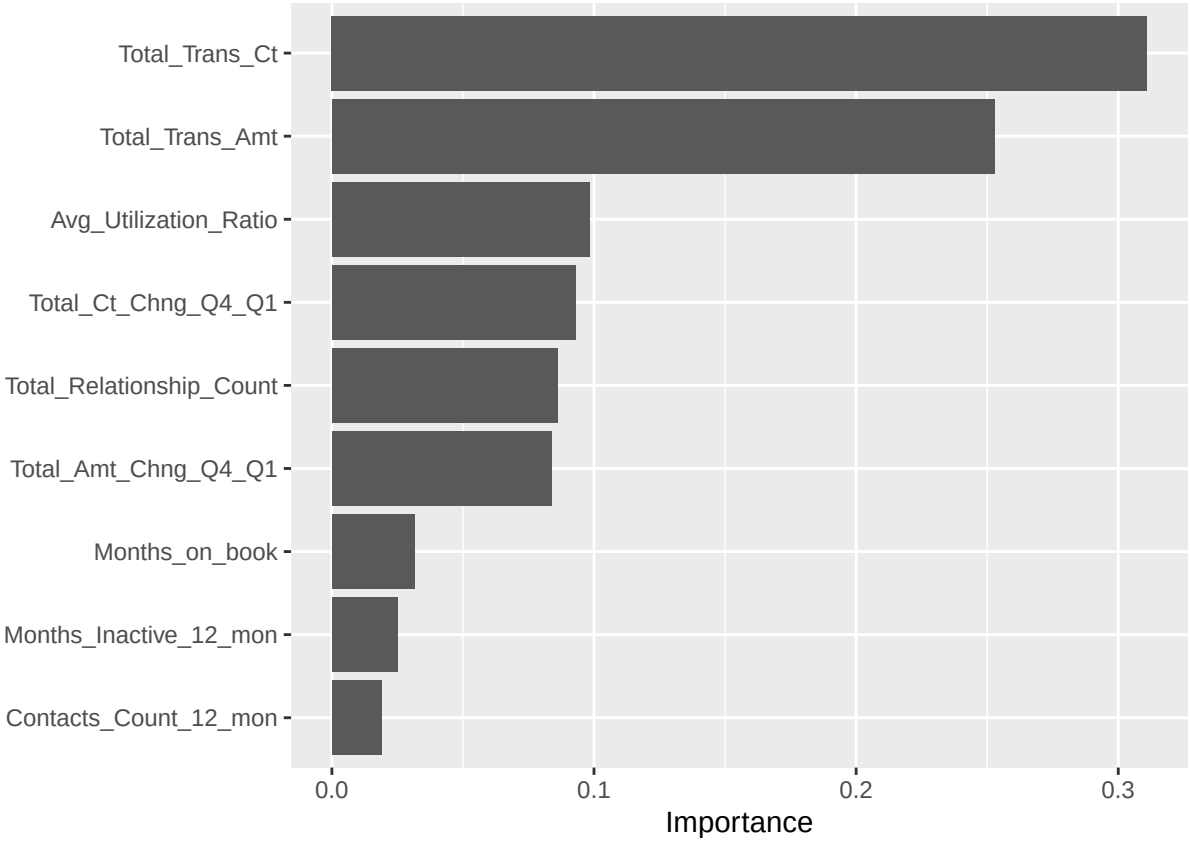
```
autoplot(xgboost_results)
```




```
best_auc <- xgboost_results %>% select_best(metric = "roc_auc")
best_acc <- xgboost_results %>% select_best(metric = "accuracy")
```

```
########################## use best hyperparameter values ##################################
final_xgboost_workflow <- finalize_workflow(
  xgboost_workflow,
  best_auc
)
# ^ this output isn't readable, but its valid

last_xgboost_fit <-
  final_xgboost_workflow %>%
  last_fit(data_split)

xgboost_metrics <- last_xgboost_fit %>% collect_metrics()

# ^ last_fit will train on train split and validate on test split w/o requiring specific hardcoding in
```

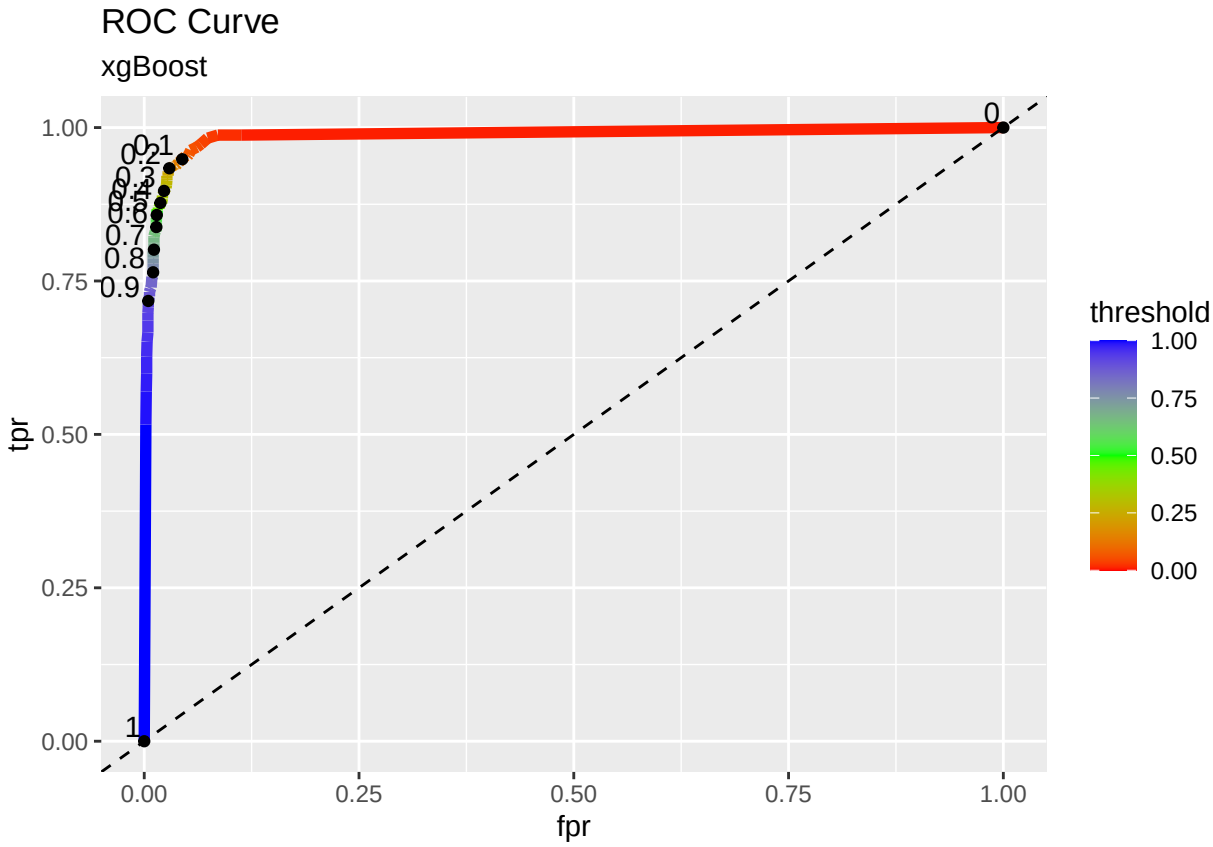
```
################################### feature importance ######################################

last_xgboost_fit %>%
  extract_fit_parsnip() %>%
  vip(num_features = 10)  # check this number
```



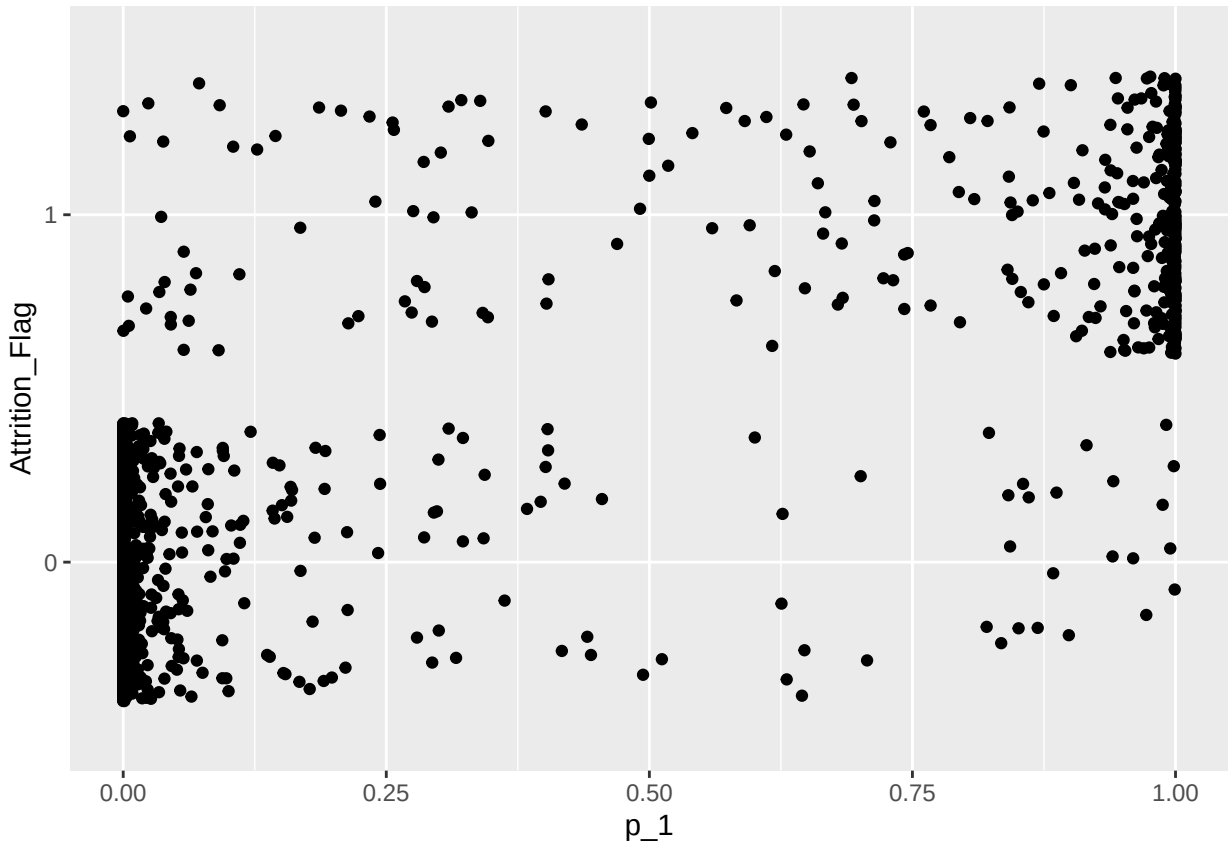

```
# compare to the null model
null_classification <- null_model() %>%
  set_engine("parsnip")%>%
  set_mode("classification")

null_xb <- workflow() %>%
  add_recipe(attrition_recipe_xb) %>%
  add_model(null_classification) %>%
  fit_resamples(
    cv_set
  )

null_metrics <- null_xb %>% collect_metrics()
```

```
################################### plot ROC curve ###################################
# same as other times we've plotted curve, substituting the xgboost
test_results <-
  test_set %>%
  select(Attrition_Flag) %>%
  bind_cols(
    last_xgboost_fit %>% collect_predictions() %>%
      select(p_1 = .pred_1)
  )

roc_data <- data.frame(threshold=seq(1,0,-0.01), fpr=0, tpr=0)
for (i in roc_data$threshold) {
```

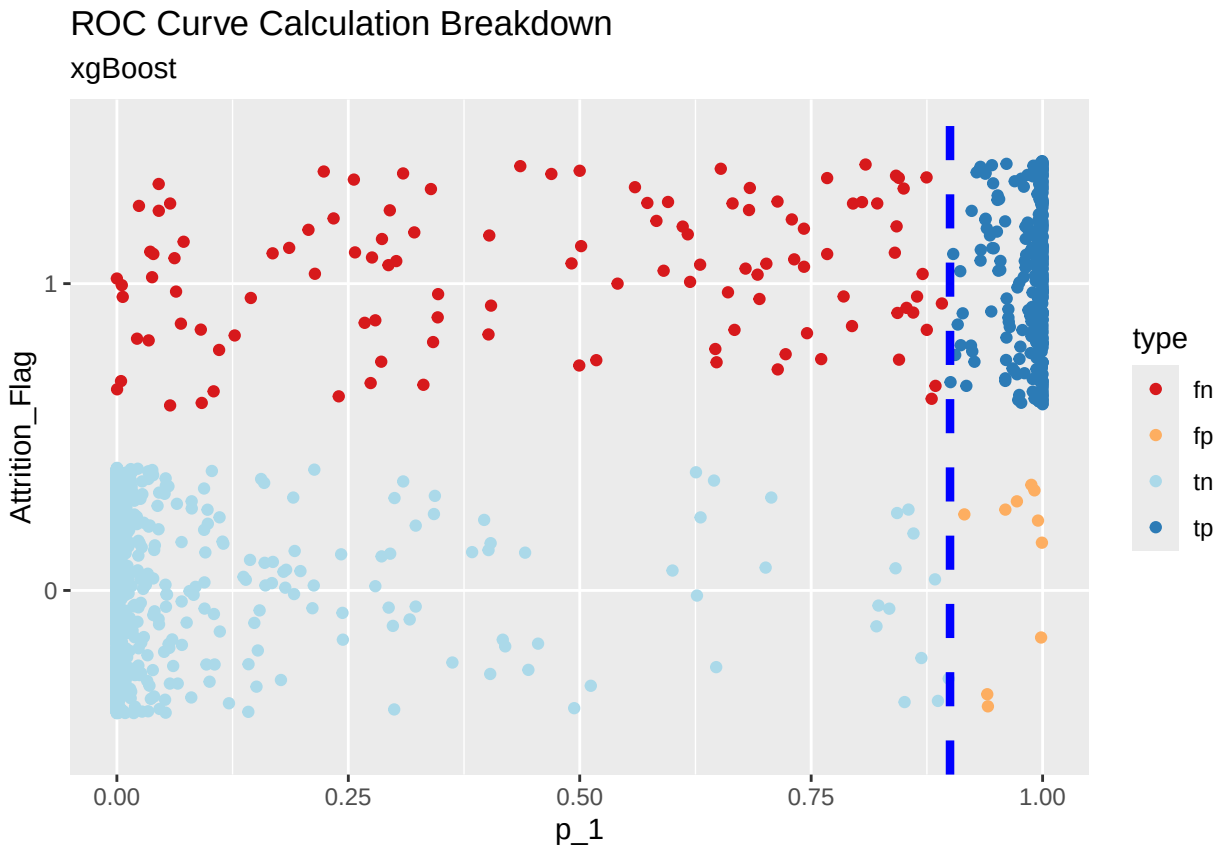
```
  over_threshold <- test_results[test_results$p_1 >= i, ]

  fpr <- sum(over_threshold$Attrition_Flag==0)/sum(test_results$Attrition_Flag==0)
  roc_data[roc_data$threshold==i, "fpr"] <- fpr

  tpr <- sum(over_threshold$Attrition_Flag==1)/sum(test_results$Attrition_Flag==1)
  roc_data[roc_data$threshold==i, "tpr"] <- tpr

}

ggplot() +
  geom_line(data = roc_data, aes(x = fpr, y = tpr, color = threshold), linewidth = 2) +
  scale_color_gradientn(colors = rainbow(3)) +
  geom_abline(intercept = 0, slope = 1, lty = 2) +
  geom_point(data = roc_data[seq(1, 101, 10), ], aes(x = fpr, y = tpr)) +
  geom_text(data = roc_data[seq(1, 101, 10), ],
            aes(x = fpr, y = tpr, label = threshold, hjust = 1.2, vjust = -0.2)) +
  ggtitle("ROC Curve", subtitle = "xgBoost")
```




```
############################## ROC curve calculation breakdown ############################

ggplot(data = test_results, aes(x = p_1, y = Attrition_Flag)) +
  geom_jitter()
```




```
threshold <- 0.9

test_results$predictions <- ifelse(test_results$p_1 >= threshold, 1, 0)
tp <- nrow(test_results[test_results$Attrition_Flag==1 & test_results$predictions==1, ])
fp <- nrow(test_results[test_results$Attrition_Flag==0 & test_results$predictions==1, ])
tn <- nrow(test_results[test_results$Attrition_Flag==0 & test_results$predictions==0, ])
fn <- nrow(test_results[test_results$Attrition_Flag==1 & test_results$predictions==0, ])

test_results$type <- ""
test_results[test_results$Attrition_Flag==1 & test_results$predictions==1, "type"] <- "tp"
test_results[test_results$Attrition_Flag==0 & test_results$predictions==1, "type"] <- "fp"
test_results[test_results$Attrition_Flag==0 & test_results$predictions==0, "type"] <- "tn"
test_results[test_results$Attrition_Flag==1 & test_results$predictions==0, "type"] <- "fn"

ggplot(data = test_results, aes(x = p_1, y = Attrition_Flag)) +
  geom_jitter(aes(colour = type)) +
  geom_vline(xintercept = threshold, linetype = "dashed", color = "blue", linewidth = 1.5) +
  scale_color_brewer(palette = "RdYlBu") +
  ggtitle("ROC Curve Calculation Breakdown", subtitle = "xgBoost")
```




```
fpr <- fp/(fp + tn)
tpr <- tp/(tp + fn)

# confusion matrix data
tn
```

```
## [1] 2115
```

```
fn
```

```
## [1] 115
```

```
fp
```

```
## [1] 10
```

```
tp
```

```
## [1] 292
```